Within-cluster substrate chemistry, colored by dominant protein type (77 of 82 clusters, sorted by me

Glucocorticoid receptor (n_prot=191, n_sub=2)
Glucosylceramidase (n_prot=324, n_sub=4)
Microsomal triglyceride transfer protein large subunit (n_prot=9, n_sub=3)
Hedgehog protein (n_prot=224, n_sub=2)
Hormone-sensitive lipase (n_prot=220, n_sub=3)
Apolipoprotein A-I (n_prot=59, n_sub=2)
Hedgehog protein (n_prot=212, n_sub=2)
Bile acid receptor (n_prot=58, n_sub=2)
Acyl-coenzyme A diphosphatase NUDT19 (n_prot=205, n_sub=2)
Glucosylceramidase (n_prot=186, n_sub=7)
Bile salt export pump (n_prot=83, n_sub=10)
Multidrug resistance-associated protein 4 (n_prot=103, n_sub=9)
G-protein coupled estrogen receptor 1 (n_prot=159, n_sub=10)
Acyl-coenzyme A diphosphatase NUDT19 (n_prot=252, n_sub=17)
Hormone-sensitive lipase (n_prot=383, n_sub=8)
Non-lysosomal glucosylceramidase (n_prot=112, n_sub=7)
Non-lysosomal glucosylceramidase (n_prot=90, n_sub=7)
Lanosterol 14-alpha demethylase (n_prot=164, n_sub=12)
Cytochrome P450 85A1 (n_prot=11, n_sub=29)
Cholesterol 24-hydroxylase (n_prot=200, n_sub=20)
Caveolin (n_prot=77, n_sub=12)
Broad substrate specificity ATP-binding cassette transp (n_prot=265, n_sub=3)
Broad substrate specificity ATP-binding cassette transp (n_prot=342, n_sub=3)
Broad substrate specificity ATP-binding cassette transp (n_prot=166, n_sub=3)
Non-lysosomal glucosylceramidase (n_prot=50, n_sub=10)
Sterol O-acyltransferase 1 (n_prot=46, n_sub=61)
Glucosylceramidase (n_prot=385, n_sub=12)
Multidrug resistance-associated protein 1 (n_prot=40, n_sub=16)
Cholesterol 24-hydroxylase (n_prot=713, n_sub=26)
Non-lysosomal glucosylceramidase (n_prot=68, n_sub=20)
Aldo-keto reductase family 1 member C1 (n_prot=148, n_sub=42)
Peroxisomal coenzyme A diphosphatase NUDT7 (n_prot=388, n_sub=13)
Delta(24)-sterol reductase (n_prot=41, n_sub=58)
Sterol carrier protein 2 (n_prot=311, n_sub=50)
Glucosylceramidase (n_prot=46, n_sub=11)
Phosphatidylcholine-sterol acyltransferase (n_prot=42, n_sub=38)
Apolipoprotein A-II (n_prot=83, n_sub=23)
Patched 1 (n_prot=89, n_sub=23)
Sterol carrier protein 2 (n_prot=69, n_sub=58)
Carboxylic ester hydrolase (n_prot=563, n_sub=3)
(3R)-3-hydroxyacyl-CoA dehydrogenase (n_prot=67, n_sub=26)
3-keto-steroid reductase (n_prot=104, n_sub=17)
Peroxisomal multifunctional enzyme type 2 (n_prot=82, n_sub=34)
Multidrug resistance-associated protein 1 (n_prot=43, n_sub=45)
Multidrug resistance-associated protein 1 (n_prot=35, n_sub=36)
Cytochrome P450 1A (n_prot=445, n_sub=26)
3-hydroxyacyl-CoA dehydrogenase type-2 (n_prot=313, n_sub=24)
3-hydroxyacyl-CoA dehydrogenase type-2 (n_prot=168, n_sub=44)
Androgen receptor (n_prot=105, n_sub=14)
Non-lysosomal glucosylceramidase (n_prot=82, n_sub=42)
Cholesterol 24-hydroxylase (n_prot=199, n_sub=66)
11-beta-hydroxysteroid dehydrogenase 1 (n_prot=252, n_sub=30)
Sterol carrier protein 2 (n_prot=381, n_sub=43)
Organic solute transporter subunit alpha (n_prot=33, n_sub=47)
Steroid 17-alpha-hydroxylase (n_prot=144, n_sub=25)
Carboxylic ester hydrolase (n_prot=198, n_sub=99)
Multidrug resistance-associated protein 1 (n_prot=81, n_sub=20)
Cholesterol 24-hydroxylase (n_prot=59, n_sub=64)
24-hydroxycholesterol 7-alpha-hydroxylase (n_prot=268, n_sub=39)
Acyl-coenzyme A thioesterase 8 (n_prot=572, n_sub=17)
24-hydroxycholesterol 7-alpha-hydroxylase (n_prot=56, n_sub=38)
Sterol carrier protein 2 (n_prot=496, n_sub=19)
Cholesterol 24-hydroxylase (n_prot=182, n_sub=44)
Carboxylic ester hydrolase (n_prot=368, n_sub=11)
Cytochrome P450 3A (n_prot=73, n_sub=42)
3-hydroxyacyl-CoA dehydrogenase type-2 (n_prot=63, n_sub=108)
Oxysterol-binding protein-related protein 8 (n_prot=110, n_sub=65)
11-beta-hydroxysteroid dehydrogenase 1 (n_prot=204, n_sub=87)
Non-lysosomal glucosylceramidase (n_prot=181, n_sub=30)
Hormone-sensitive lipase (n_prot=104, n_sub=22)
Hormone-sensitive lipase (n_prot=299, n_sub=39)
Glucosylceramidase (n_prot=281, n_sub=20)
Glucosylceramidase (n_prot=93, n_sub=20)
Sterol carrier protein 2 (n_prot=310, n_sub=5)
Bile acid receptor (n_prot=82, n_sub=29)
G-protein coupled estrogen receptor 1 (n_prot=66, n_sub=27)
(3R)-3-hydroxyacyl-CoA dehydrogenase (n_prot=16, n_sub=17)

Enzyme
Receptor
Transporter
Other
Mixed
Random-pair Tanimoto median = 0.20

0.0
0.2
0.4
0.6
0.8
1.0

Pairwise Tanimoto (ECFP4) — within cluster